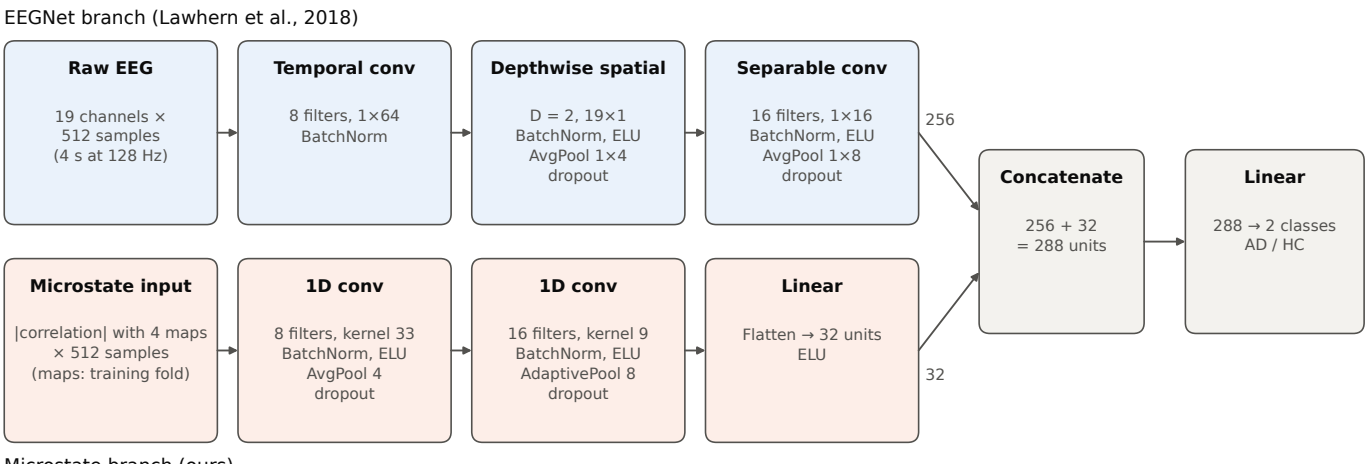
EEGNet branch (Lawhern et al., 2018)
Raw EEG
19 channels ×
512 samples
(4 s at 128 Hz)
Temporal conv
8 filters, 1×64
BatchNorm
Depthwise spatial
D = 2, 19×1
BatchNorm, ELU
AvgPool 1×4
dropout
Separable conv
16 filters, 1×16
BatchNorm, ELU
AvgPool 1×8
dropout
256
Concatenate
256 + 32
= 288 units
Linear
288 → 2 classes
AD / HC
Microstate input
|correlation| with 4 maps
× 512 samples
(maps: training fold)
1D conv
8 filters, kernel 33
BatchNorm, ELU
AvgPool 4
dropout
1D conv
16 filters, kernel 9
BatchNorm, ELU
AdaptivePool 8
dropout
Linear
Flatten → 32 units
ELU
32
Microstate branch (ours)